MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. The angle of elevation of the top of a tower at a point on the ground 50 m away from the foot of the tower is 45°. Then the height of the tower (in metres) is: [1]

(A) 25 m (B) 50 m (C) $50\sqrt{3}$ m (D) $\dfrac{50}{\sqrt{3}}$ m

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B)
Justification: $\tan 45^\circ = \dfrac{h}{50} \implies 1 = \dfrac{h}{50} \implies h = \boxed{50 \text{ m}}$.

Q2. The angle of depression of a car, standing on the ground, from the top of a 75 m high tower is 30°. The distance of the car from the base of the tower (in metres) is: [1]

(A) $25\sqrt{3}$ (B) $50\sqrt{3}$ (C) $75\sqrt{3}$ (D) 150

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C)
Justification: Alternate angle of elevation is 30°. Thus $\tan 30^\circ = \dfrac{75}{d} \implies d = \boxed{75\sqrt{3} \text{ m}}$.

Q3. The angle of depression of a car parked on the road from the top of a 150 m high cliff is 30°. The distance of the car from the base of the cliff (in metres) is: [1]

(A) $50\sqrt{3}$ (B) $150\sqrt{3}$ (C) 150 (D) 75

► **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B)
Justification: By alternate interior angles, elevation at car = 30°. Hence $d = 150 \cot 30^\circ = \boxed{150\sqrt{3} \text{ m}}$.

Q4. A ladder 15 m long just reaches the top of a vertical wall. If the ladder makes an angle of 60° with the wall, then the height of the wall is: [1]

(A) $15\sqrt{3}$ m (B) $\frac{15\sqrt{3}}{2}$ m (C) 7.5 m (D) 15 m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: The 60° angle is with the vertical wall (adjacent side = h), so $\cos 60° = \frac{h}{15} \implies h = 15 \times \frac{1}{2} = \boxed{7.5 \text{ m}}$.

Q5. Assertion (A): When the angle of elevation of the sun increases from 30° to 60°, the length of the shadow of a vertical tower decreases.
Reason (R): For a fixed height h, shadow length $s = h\cot\theta$, and $\cot\theta$ is a strictly decreasing function for $0° < \theta < 90°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: As θ rises from 30° to 60°, $\cot\theta$ decreases from $\sqrt{3}$ to $\frac{1}{\sqrt{3}}$, so $s = h\cot\theta$ strictly decreases. Both (A) and (R) are true and (R) is the correct explanation of (A).

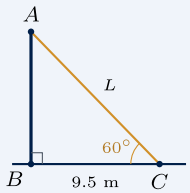
SECTION B: Short Answer Questions

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Q6. A ladder leaning against a vertical wall makes an angle of 60° with the horizontal ground. If the foot of the ladder is 9.5 m away from the wall, calculate the length of the ladder. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let AC be the ladder leaning against wall AB, with $BC = 9.5$ m and $\angle ACB = 60°$. [0.5 Mark]



In right $\triangle ABC$ ($\angle B = 90°$):

$$\cos 60° = \frac{BC}{AC} \implies \frac{1}{2} = \frac{9.5}{AC}$$

Applying $\cos 60°$: [1 Mark]

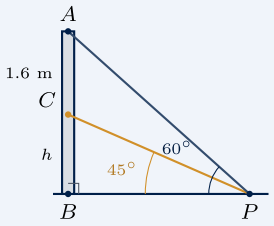
$$\implies AC = 2 \times 9.5 = \boxed{19 \text{ m}}$$

Final length of ladder: [0.5 Mark]

Q7. A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45°. Find the height of the pedestal. [Take $\sqrt{3} = 1.732$] **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let $BC = h$ m be the pedestal and $AC = 1.6$ m be the statue above it, observed from point P on ground with $\angle BPC = 45°$ and $\angle BPA = 60°$. [0.5 Mark]



In right $\triangle CBP$:

$$\tan 45° = \frac{BC}{BP} \implies 1 = \frac{h}{BP} \implies BP = h$$

Finding $BP = h$: [1 Mark]

In right $\triangle ABP$:

$$\tan 60° = \frac{AB}{BP} \implies \sqrt{3} = \frac{h + 1.6}{h}$$

Applying $\tan 60°$: [0.5 Mark]

$$h\sqrt{3} = h + 1.6 \implies h(\sqrt{3} - 1) = 1.6 \implies h = \frac{1.6}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{1.6(\sqrt{3} + 1)}{2} = 0.8(\sqrt{3} + 1) \text{ m.}$$

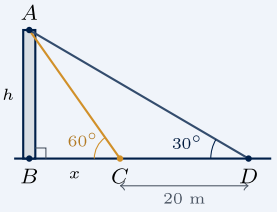
[0.5 Mark]

Using $\sqrt{3} = 1.732$: $h = 0.8(1.732 + 1) = 0.8 \times 2.732 = \boxed{2.1856 \text{ m} \approx 2.19 \text{ m}}$. [0.5 Mark]

Q8. A TV tower stands on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower and the width of the canal. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$ m, canal width $BC = x$ m, and outer point D have $CD = 20$ m, with $\angle ACB = 60°$ and $\angle ADB = 30°$. [0.5 Mark]



In right $\triangle ABC$: $\tan 60° = \frac{h}{x} \implies h = \sqrt{3}\,x$ — (i). In right $\triangle ABD$: $\tan 30° = \frac{h}{x + 20} \implies h = \frac{x + 20}{\sqrt{3}}$ — (ii). [1 Mark]

Equating (i) and (ii): $\sqrt{3}\,x = \frac{x + 20}{\sqrt{3}} \implies 3x = x + 20 \implies 2x = 20 \implies x = 10$ m. [1 Mark]

Hence $h = 10\sqrt{3} = 10 \times 1.732 = 17.32$ m.

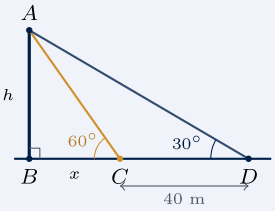
$\therefore$ Width of canal $= \boxed{10 \text{ m}}$, Height of tower $= \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$

Final values: [0.5 Mark]

Q9. The shadow of a vertical tower on level ground is found to be 40 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$ m, shadow at 60° be $BC = x$ m, and shadow at 30° be $BD = (x + 40)$ m ($CD = 40$ m). **[0.5 Mark]**



In right $\triangle ABC$: $\tan 60° = \dfrac{h}{x} \implies x = \dfrac{h}{\sqrt{3}}$ — (i). In right $\triangle ABD$: $\tan 30° = \dfrac{h}{x+40} \implies x + 40 = h\sqrt{3}$ — (ii). **[1 Mark]**

Substituting (i) into (ii): $\dfrac{h}{\sqrt{3}} + 40 = h\sqrt{3} \implies 40 = h\left(\sqrt{3} - \dfrac{1}{\sqrt{3}}\right) = \dfrac{2h}{\sqrt{3}}$. **[1 Mark]**

$$\implies h = \frac{40\sqrt{3}}{2} = 20\sqrt{3} = 20 \times 1.732 = \boxed{20\sqrt{3} \text{ m} \approx 34.64 \text{ m}}$$

Final tower height: **[0.5 Mark]**

SECTION C: Applied Case Study

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Q10. CASE STUDY: Power Utility Infrastructure Maintenance

An electrical technician repairs overhead pole AB of height 6 m. For safety clearance from live conductors at A, the technician must reach working platform D located 1.5 m below A. An insulated ladder CD is inclined at 60° to the horizontal ground.

(i) What is the vertical height from the ground to the repair position point D? **[1]**

(ii) What should be the minimum length of the ladder CD required to reach point D? [Take $\sqrt{3} = 1.732$] **[1]**

(iii) How far from the foot of the pole (B) should the foot of the ladder (C) be placed on the ground? **[2]**

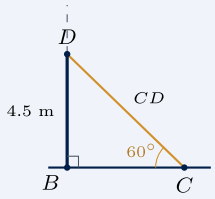
OR

If ground obstructions require the ladder to be placed at an inclination of 45° instead of 60° to reach the same height of 4.5 m, find the new required length of the ladder. **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Since pole height $AB = 6$ m and point D is 1.5 m below top A:

$$BD = AB - AD = 6 - 1.5 = \boxed{4.5 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) In right $\triangle DBC$ ($\angle DCB = 60°$):

$$\sin 60° = \frac{BD}{CD} \implies \frac{\sqrt{3}}{2} = \frac{4.5}{CD}$$

$$\implies CD = \frac{9}{\sqrt{3}} = 3\sqrt{3} = 3(1.732) = \boxed{5.196 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: In right $\triangle DBC$, $\tan 60° = \frac{BD}{BC} \implies \sqrt{3} = \frac{4.5}{BC}$. [1 Mark]

$$BC = \frac{4.5}{\sqrt{3}} = 1.5\sqrt{3} = 1.5(1.732) = \boxed{2.598 \text{ m} \approx 2.60 \text{ m}}$$

Subpart (iii) distance BC: [1 Mark]

(iii) OR Choice: At inclination 45° to reach height $BD = 4.5$ m, $\sin 45° = \frac{4.5}{CD'}$. [1 Mark]

$$\implies CD' = 4.5\sqrt{2} \approx 4.5 \times 1.414 = \boxed{4.5\sqrt{2} \text{ m} \approx 6.36 \text{ m}}$$

New ladder length: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★